Code:-

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/stat.h>
#include <fcntl.h>

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#define FIFO1 "/tmp/fifo1"
#define FIFO2 "/tmp/fifo2"
#define BUFFER_SIZE 1024

void process1() {
    int fd_write, fd_read;
    char input_string[BUFFER_SIZE];
    char output_string[BUFFER_SIZE];

    printf("[Process 1] Enter sentences (Press Enter then Ctrl+D to finish):\n");

    // Read input from user until EOF
    char temp[BUFFER_SIZE];
    input_string[0] = '\0';
    while (fgets(temp, BUFFER_SIZE, stdin) != NULL) {
        strcat(input_string, temp);
    }

    // Open FIFO1 for writing user input
    fd_write = open(FIFO1, O_WRONLY);
    write(fd_write, input_string, strlen(input_string) + 1);
    close(fd_write);

    // Open FIFO2 for reading the processed results
    fd_read = open(FIFO2, O_RDONLY);
    read(fd_read, output_string, BUFFER_SIZE);
    close(fd_read);

    // Display the received content on standard output
    printf("\n[Process 1] Received Results from Process 2:\n%s", output_string);
}

void process2() {
```

```
int fd_read, fd_write;
char buffer[BUFFER_SIZE];
int characters = 0, words = 0, lines = 0;
FILE *file;

// Open FIFO1 to read the sentence from Process 1
fd_read = open(FIFO1, O_RDONLY);
read(fd_read, buffer, BUFFER_SIZE);
close(fd_read);

// Count characters, words, and lines
int i = 0;
int in_word = 0;
while (buffer[i] != '\0') {
   characters++;

   if (buffer[i] == '\n') {
      lines++;
   }

   if (buffer[i] == ' ' || buffer[i] == '\t' || buffer[i] == '\n') {
      in_word = 0;
   } else if (in_word == 0) {
      in_word = 1;
      words++;
   }
   i++;
}
// If the text doesn't end with a newline but has content, count it as a line
if (characters > 0 && buffer[i-1] != '\n') {
   lines++;
}

// Write output to a text file
file = fopen("output.txt", "w");
if (file == NULL) {
   perror("Error opening file");
   exit(1);
}
fprintf(file, "Character count: %d\n", characters);
fprintf(file, "Word count: %d\n", words);
fprintf(file, "Line count: %d\n", lines);
fclose(file);

// Read the contents back from the text file
char file_content[BUFFER_SIZE];
char line[128];
file_content[0] = '\0';

file = fopen("output.txt", "r");
```

```
    while (fgets(line, sizeof(line), file) != NULL) {
        strcat(file_content, line);
    }
    fclose(file);

    // Open FIFO2 to send file contents back to Process 1
    fd_write = open(FIFO2, O_WRONLY);
    write(fd_write, file_content, strlen(file_content) + 1);
    close(fd_write);
}

int main() {
    // Create the named pipes (FIFOs)
    mkfifo(FIFO1, 0666);
    mkfifo(FIFO2, 0666);

    pid_t pid = fork();

    if (pid < 0) {
        perror("Fork failed");
        return 1;
    }

    if (pid > 0) {
        // Parent execution block
        process1();
    } else {
        // Child execution block
        process2();
        exit(0);
    }

    // Clean up FIFOs after communication ends
    unlink(FIFO1);
    unlink(FIFO2);

    return 0;
}
```

OUTPUT:-

```
avcoe@avcoe-HP-Compaq-Pro-4300-SFF-PC:~$ gcc as6.c -o as6 && ./as6
[Process 1] Enter sentences (Press Enter then Ctrl+D to finish):
Hello World
Operating System
Nmaed Pipes Communication

[Process 1] Received Results from Process 2:
Character count: 54
Word count: 7
Line count: 3
```